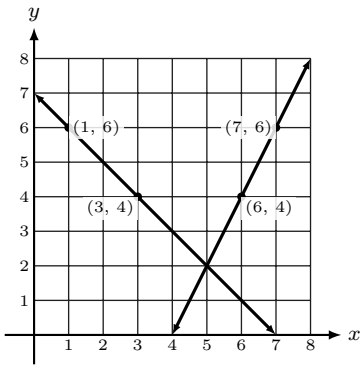
Artifact 17 – Points of Intersection

Driving Question: How do you find the point where two lines meet, verify it, and say what it means in a situation?

Example 1

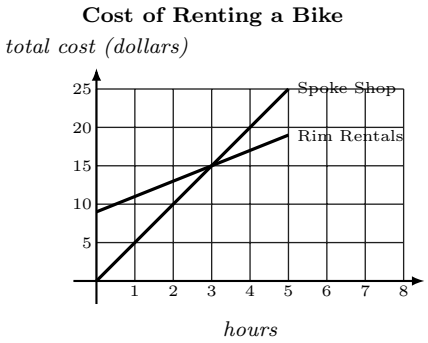


The graphs of $y = 2x - 8$ and $y = -x + 7$ are shown.

Write the coordinates of the point of intersection.

Verify your ordered pair. Show the check for each equation.

Example 2



Spoke Shop and Rim Rentals each rent bikes by the hour. The graph shows the total cost.

Write the coordinates of the point of intersection.

What does the x-coordinate represent in this situation? What does the y-coordinate represent?

Example 3

Two lines have the equations $y = 4x - 3$ and $y = x + 12$.

Determine the ordered pair where the graphs of the two equations intersect. Show your work, then show the check in both equations.

Vocabulary

point of intersection	the point where two graphed lines cross; its coordinates satisfy both equations
satisfy (an equation)	a pair of values satisfies an equation if substituting them makes the equation true
substitution	replacing x and y in an equation with values to check whether the equation is true

Notes to Forgetful Self

Given	How to find the point of intersection	How to verify	What (x, y) tells you
two lines on a grid			
a real-world graph			
two equations $y = mx + b$			

A point that is on only one line _____ a solution of the pair of equations.